

12345678

WPC POWER DRIVER BOARD – COST–OPTIMISED MODERN VERSION – functional replacement for Williams A–12697–3 (same outline, connectors, fuses).  
DPAK MOSFET solenoid/lamp drivers, TPS54360B buck regulators for +5V/+12V, 4–quadrant insulated triacs, SMD passives, 5x20 mm fuses.  
Sheets: Power\_Supply, CPU\_Interface, Sol\_HighPower (1–8), Sol\_LowPower (9–16), Sol\_Flashers (17–24), Sol\_GeneralPurpose (25–28),  
Lamp\_Columns, Lamp\_Rows, GI\_Triacs. Every part carries Manufacturer / MPN / DigiKey link / Price / Price100 fields (see BOM). See README.md.

Power\_Supply

File: power\_supply.kicad\_sch

CPU\_Interface

File: cpu\_interface.kicad\_sch

Sol\_HighPower

File: sol\_highpower.kicad\_sch

Sol\_LowPower

File: sol\_lowpower.kicad\_sch

Sol\_Flashers

File: sol\_flashers.kicad\_sch

Sol\_GeneralPurpose

File: sol\_gp.kicad\_sch

Lamp\_Columns

File: lamp\_columns.kicad\_sch

Lamp\_Rows

File: lamp\_rows.kicad\_sch

GI\_Triacs

File: gi\_triacs.kicad\_sch

Modern re–implementation of the Williams A–12697–3 (same form factor)

Sheet: /  
File: wpc\_power\_driver\_cost.kicad\_sch

Title: WPC Power Driver Board – cost–optimised modern re–implementation

Size: A3Date: 2026–09–04Rev: A

KiCad E.D.A. 10.0.6Id: 1/10

12345678



[illegible]

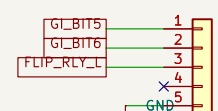
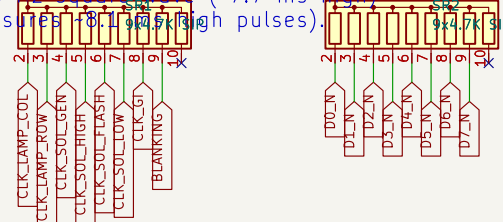


CPU interface, pin-out from the WPC schematic manual 16-9834.2. J113: 7 /LMP ROW, 8 /LMP COL, 9 /SOL 2 (sol 17-24), 10 /SOL 4 (sol 1-8), 11 /SOL 3 (sol 9-16), 12 /SOL 1 (sol 25-28), 13 /TRIAC, 15, 22, 23 odd = /D7../D0, 31 BLANKING (HIGH = every latch tri-stated, open-collector from the CPU, pulled up here), 34 Z800 CROSS to the CPU, 1 /V, 6 /V, 12, 27 odd = every latch tri-stated, open-collector from the CPU, pulled up here, 4 /V, 5 /V, 11 /V, 13 /V, 14 /V, 15 /V, 16 /V, 17 /V, 18 /V, 19 /V, 20 /V, 21 /V, 22 /V, 23 /V, 24 /V, 25 /V, 26 /V, 27 /V, 28 /V, 29 /V, 30 /V, 31 /V, 32 /V, 33 /V, 34 /V, 35 /V, 36 /V, 37 /V, 38 /V, 39 /V, 40 /V, 41 /V, 42 /V, 43 /V, 44 /V, 45 /V, 46 /V, 47 /V, 48 /V, 49 /V, 50 /V, 51 /V, 52 /V, 53 /V, 54 /V, 55 /V, 56 /V, 57 /V, 58 /V, 59 /V, 60 /V, 61 /V, 62 /V, 63 /V, 64 /V, 65 /V, 66 /V, 67 /V, 68 /V, 69 /V, 70 /V, 71 /V, 72 /V, 73 /V, 74 /V, 75 /V, 76 /V, 77 /V, 78 /V, 79 /V, 80 /V, 81 /V, 82 /V, 83 /V, 84 /V, 85 /V, 86 /V, 87 /V, 88 /V, 89 /V, 90 /V, 91 /V, 92 /V, 93 /V, 94 /V, 95 /V, 96 /V, 97 /V, 98 /V, 99 /V, 100 /V, 101 /V, 102 /V, 103 /V, 104 /V, 105 /V, 106 /V, 107 /V, 108 /V, 109 /V, 110 /V, 111 /V, 112 /V, 113 /V, 114 /V, 115 /V, 116 /V, 117 /V, 118 /V, 119 /V, 120 /V, 121 /V, 122 /V, 123 /V, 124 /V, 125 /V, 126 /V, 127 /V, 128 /V, 129 /V, 130 /V, 131 /V, 132 /V, 133 /V, 134 /V, 135 /V, 136 /V, 137 /V, 138 /V, 139 /V, 140 /V, 141 /V, 142 /V, 143 /V, 144 /V, 145 /V, 146 /V, 147 /V, 148 /V, 149 /V, 150 /V, 151 /V, 152 /V, 153 /V, 154 /V, 155 /V, 156 /V, 157 /V, 158 /V, 159 /V, 160 /V, 161 /V, 162 /V, 163 /V, 164 /V, 165 /V, 166 /V, 167 /V, 168 /V, 169 /V, 170 /V, 171 /V, 172 /V, 173 /V, 174 /V, 175 /V, 176 /V, 177 /V, 178 /V, 179 /V, 180 /V, 181 /V, 182 /V, 183 /V, 184 /V, 185 /V, 186 /V, 187 /V, 188 /V, 189 /V, 190 /V, 191 /V, 192 /V, 193 /V, 194 /V, 195 /V, 196 /V, 197 /V, 198 /V, 199 /V, 200 /V, 201 /V, 202 /V, 203 /V, 204 /V, 205 /V, 206 /V, 207 /V, 208 /V, 209 /V, 210 /V, 211 /V, 212 /V, 213 /V, 214 /V, 215 /V, 216 /V, 217 /V, 218 /V, 219 /V, 220 /V, 221 /V, 222 /V, 223 /V, 224 /V, 225 /V, 226 /V, 227 /V, 228 /V, 229 /V, 230 /V, 231 /V, 232 /V, 233 /V, 234 /V, 235 /V, 236 /V, 237 /V, 238 /V, 239 /V, 240 /V, 241 /V, 242 /V, 243 /V, 244 /V, 245 /V, 246 /V, 247 /V, 248 /V, 249 /V, 250 /V, 251 /V, 252 /V, 253 /V, 254 /V, 255 /V, 256 /V, 257 /V, 258 /V, 259 /V, 260 /V, 261 /V, 262 /V, 263 /V, 264 /V, 265 /V, 266 /V, 267 /V, 268 /V, 269 /V, 270 /V, 271 /V, 272 /V, 273 /V, 274 /V, 275 /V, 276 /V, 277 /V, 278 /V, 279 /V, 280 /V, 281 /V, 282 /V, 283 /V, 284 /V, 285 /V, 286 /V, 287 /V, 288 /V, 289 /V, 290 /V, 291 /V, 292 /V, 293 /V, 294 /V, 295 /V, 296 /V, 297 /V, 298 /V, 299 /V, 300 /V, 301 /V, 302 /V, 303 /V, 304 /V, 305 /V, 306 /V, 307 /V, 308 /V, 309 /V, 310 /V, 311 /V, 312 /V, 313 /V, 314 /V, 315 /V, 316 /V, 317 /V, 318 /V, 319 /V, 320 /V, 321 /V, 322 /V, 323 /V, 324 /V, 325 /V, 326 /V, 327 /V, 328 /V, 329 /V, 330 /V, 331 /V, 332 /V, 333 /V, 334 /V, 335 /V, 336 /V, 337 /V, 338 /V, 339 /V, 340 /V, 341 /V, 342 /V, 343 /V, 344 /V, 345 /V, 346 /V, 347 /V, 348 /V, 349 /V, 350 /V, 351 /V, 352 /V, 353 /V, 354 /V, 355 /V, 356 /V, 357 /V, 358 /V, 359 /V, 360 /V, 361 /V, 362 /V, 363 /V, 364 /V, 365 /V, 366 /V, 367 /V, 368 /V, 369 /V, 370 /V, 371 /V, 372 /V, 373 /V, 374 /V, 375 /V, 376 /V, 377 /V, 378 /V, 379 /V, 380 /V, 381 /V, 382 /V, 383 /V, 384 /V, 385 /V, 386 /V, 387 /V, 388 /V, 389 /V, 390 /V, 391 /V, 392 /V, 393 /V, 394 /V, 395 /V, 396 /V, 397 /V, 398 /V, 399 /V, 400 /V, 401 /V, 402 /V, 403 /V, 404 /V, 405 /V, 406 /V, 407 /V, 408 /V, 409 /V, 410 /V, 411 /V, 412 /V, 413 /V, 414 /V, 415 /V, 416 /V, 417 /V, 418 /V, 419 /V, 420 /V, 421 /V, 422 /V, 423 /V, 424 /V, 425 /V, 426 /V, 427 /V, 428 /V, 429 /V, 430 /V, 431 /V, 432 /V, 433 /V, 434 /V, 435 /V, 436 /V, 437 /V, 438 /V, 439 /V, 440 /V, 441 /V, 442 /V, 443 /V, 444 /V, 445 /V, 446 /V, 447 /V, 448 /V, 449 /V, 450 /V, 451 /V, 452 /V, 453 /V, 454 /V, 455 /V, 456 /V, 457 /V, 458 /V, 459 /V, 460 /V, 461 /V, 462 /V, 463 /V, 464 /V, 465 /V, 466 /V, 467 /V, 468 /V, 469 /V, 470 /V, 471 /V, 472 /V, 473 /V, 474 /V, 475 /V, 476 /V, 477 /V, 478 /V, 479 /V, 480 /V, 481 /V, 482 /V, 483 /V, 484 /V, 485 /V, 486 /V, 487 /V, 488 /V, 489 /V, 490 /V, 491 /V, 492 /V, 493 /V, 494 /V, 495 /V, 496 /V, 497 /V, 498 /V, 499 /V, 500 /V, 501 /V, 502 /V, 503 /V, 504 /V, 505 /V, 506 /V, 507 /V, 508 /V, 509 /V, 510 /V, 511 /V, 512 /V, 513 /V, 514 /V, 515 /V, 516 /V, 517 /V, 518 /V, 519 /V, 520 /V, 521 /V, 522 /V, 523 /V, 524 /V, 525 /V, 526 /V, 527 /V, 528 /V, 529 /V, 530 /V, 531 /V, 532 /V, 533 /V, 534 /V, 535 /V, 536 /V, 537 /V, 538 /V, 539 /V, 540 /V, 541 /V, 542 /V, 543 /V, 544 /V, 545 /V, 546 /V, 547 /V, 548 /V, 549 /V, 550 /V, 551 /V, 552 /V, 553 /V, 554 /V, 555 /V, 556 /V, 557 /V, 558 /V, 559 /V, 560 /V, 561 /V, 562 /V, 563 /V, 564 /V, 565 /V, 566 /V, 567 /V, 568 /V, 569 /V, 570 /V, 571 /V, 572 /V, 573 /V, 574 /V, 575 /V, 576 /V, 577 /V, 578

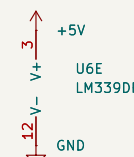
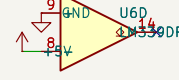
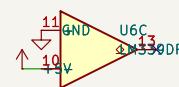
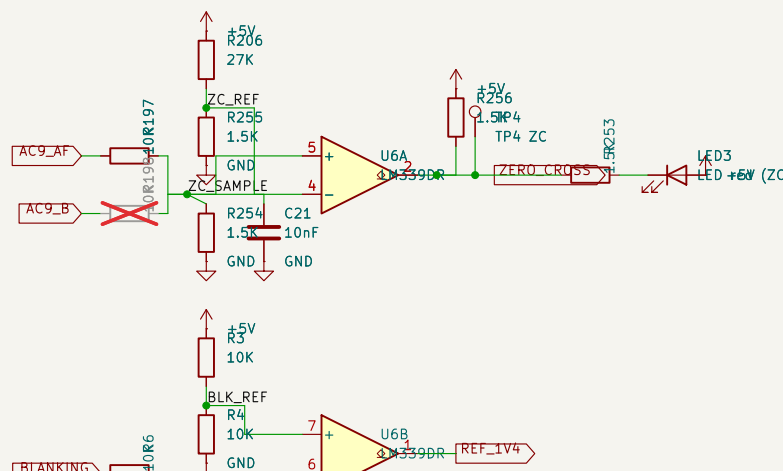
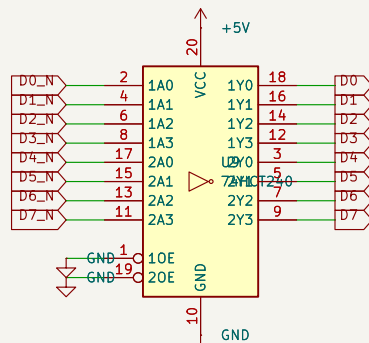
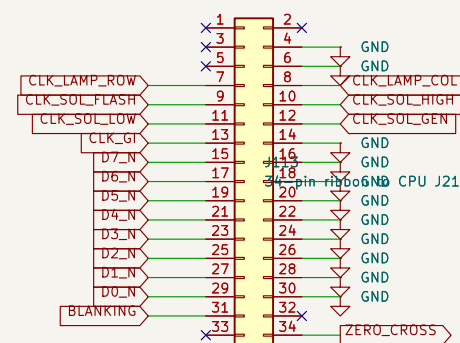
34 ZERO CROSS to the CPU; 1/3/5/32/33 (flipper-opto sense of the -1 relay board) and 2 are N/C on the A-12697-3; other even pins = ground. The CPU board drives the data through a 74LS240, so the ribbon data is ACTIVE LOW: U9 (74HCT240) inverts it and every latch then takes the

The CPU board drives the data through a 74LS240, so the ribbon data is ACTIVE LOW. 05 (74HCT240) inverts it and every latch then takes the true bus (register bit = 1 -> output ON, as PinMAME / FreeWPC write it). BLANKING tri-states every 74HCT574 (gate pull-downs -> all OFF) and pulls the lamp on; 1 / V reference low. Zero across the 5117, ground log of the 9 VAC winding is sampled through P103 10K / P254 1.5K / G31 (P108 RPN1).

the lamp-row 1.4 V reference low. Zero-cross: the F113-fused leg of the 9 VAC winding is sampled through R197 10K / R254 1.5K / C21 (R198 DNP) into the + input of U6A against the 0.26 V reference on - -> ZERO CROSS is HIGH while that leg is positive: a 60 Hz square wave (-7.7 ms high)



J111 G.I. latch spare bits T5–T7 (GPIO as on the original board, e.g. shot-clock drive; N/C on STTG



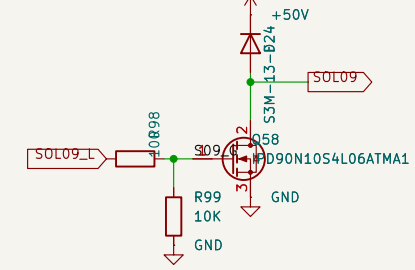




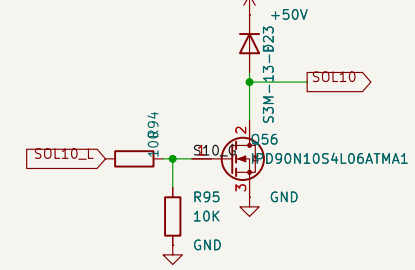


Low power solenoid drivers (sol. 9–16): IPD90N10S4L–06 logic-level DPAK MOSFETs (14 mOhm), latch U4 output HIGH = ON, clocked by /SOL 3 (J113–11). Tie-back diodes S3M (SMC, 3 A) to +50V (~70 V DC in the machine); coils in this group can be PWM-held by the firmware, the diode then carries the freewheel current. Output connectors J127 (playfield, key pin 2), J129 (backbox, sol 9–12, key 3) and J128 (cabinet, sol 13–16, key 4) as on the A–12697. Supply +50V via F104 on J107–2.

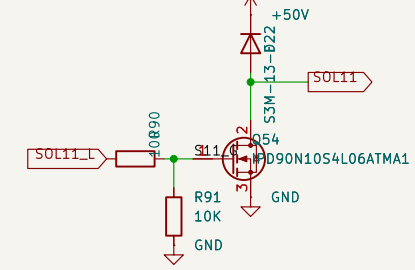
Sol 9 (low power)



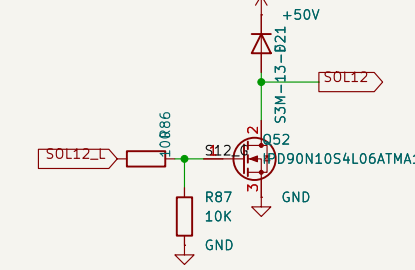
Sol 10 (low power)



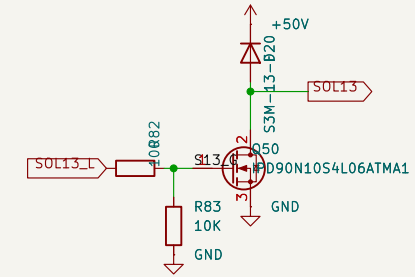
Sol 11 (low power)



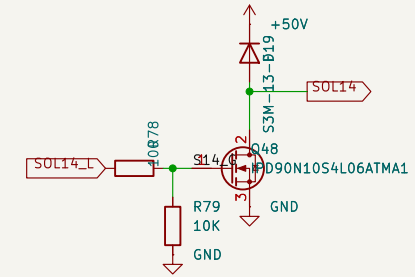
Sol 12 (low power)



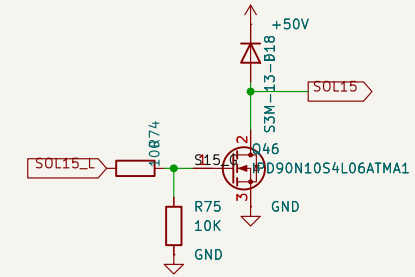
Sol 13 (low power)



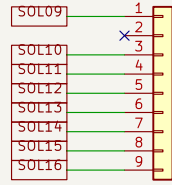
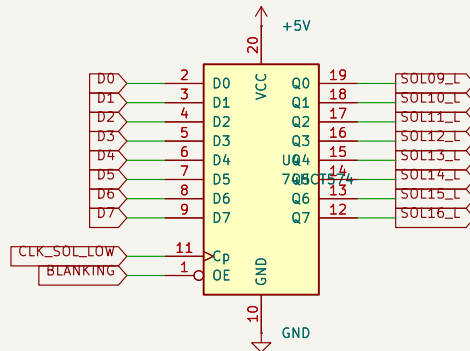
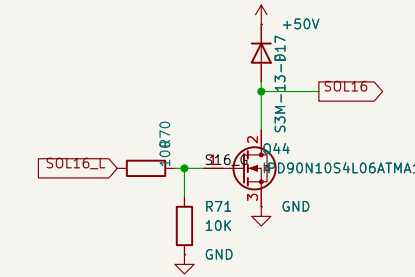
Sol 14 (low power)



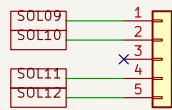
Sol 15 (low power)



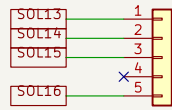
Sol 16 (low power)



J127 Sol 9–16 drive (playfield)



J129 Sol 9–12 drive (backbox, parallel to J127; N/C on STING)

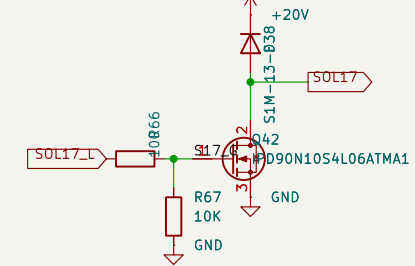


J128 Sol 13–16 drive (cabinet, parallel to J127; N/C on STING)

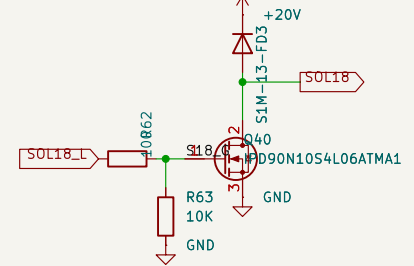


Flashlamp drivers (sol. 17-24): IPD90N10S4L-06 DPAK, latch U3 output HIGH = lamp ON, clocked by /SOL 2 (J113-9). Sol 17-20 keep an on-board S1M tie-back to +20V (the A-12697 has none there); the sol 21-24 tie-back cathodes are brought out to J126-10..13 exactly like the A-12697 D9-D12, so the harness chooses the clamp rail. On STING sol. 17/18 drive the 12 V gun motors (motor high side on the +12V power rail of J118) through this group. Outputs on J126 (playfield, key pin 9) and J125 (backbox, key pin 4). Flasher supply +20V on J107-5/6 / J106-5.

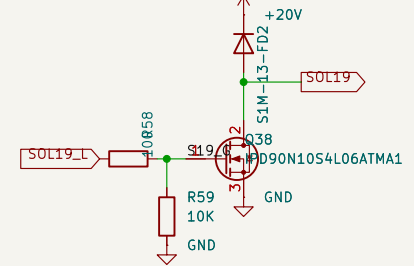
Sol 17 (flasher)



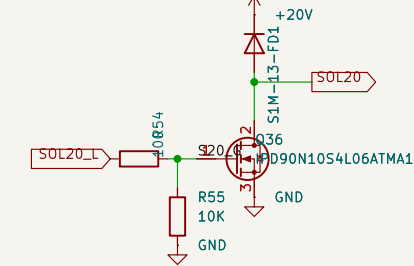
Sol 18 (flasher)



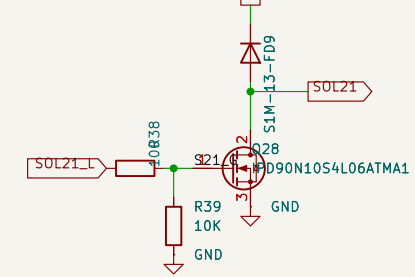
Sol 19 (flasher)



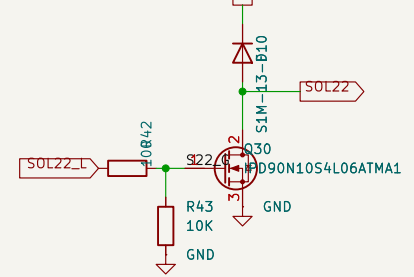
Sol 20 (flasher)



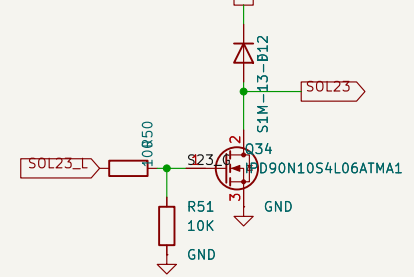
Sol 21 (flasher)



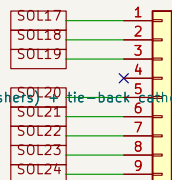
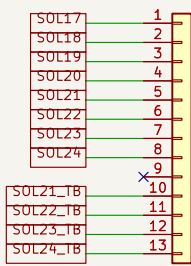
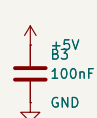
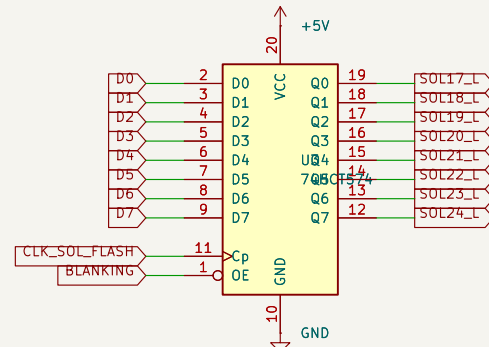
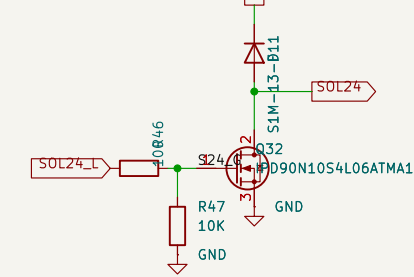
Sol 22 (flasher)



Sol 23 (flasher)

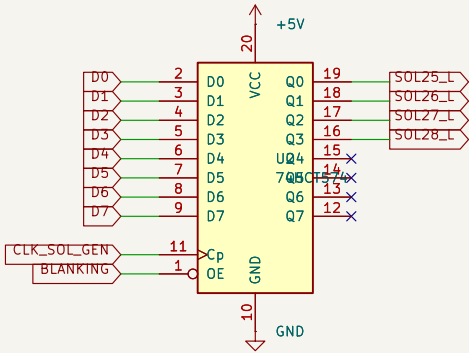
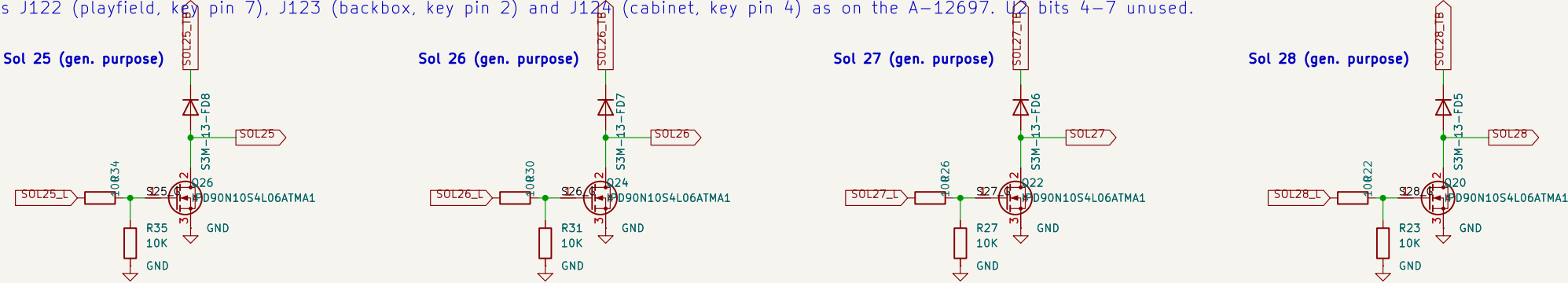


Sol 24 (flasher)



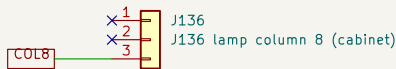
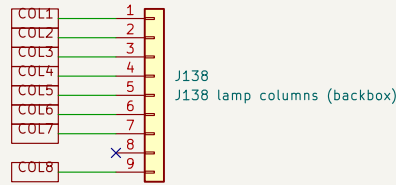
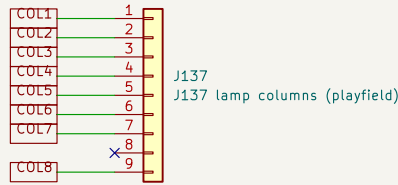
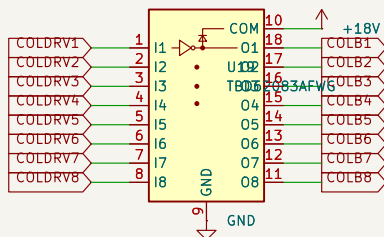
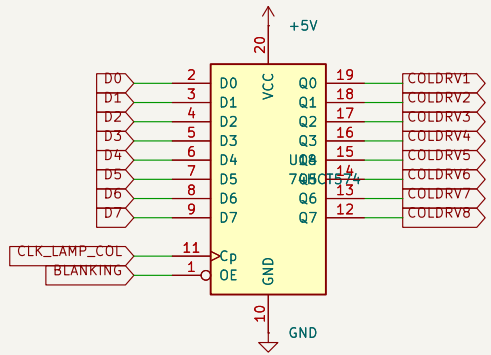
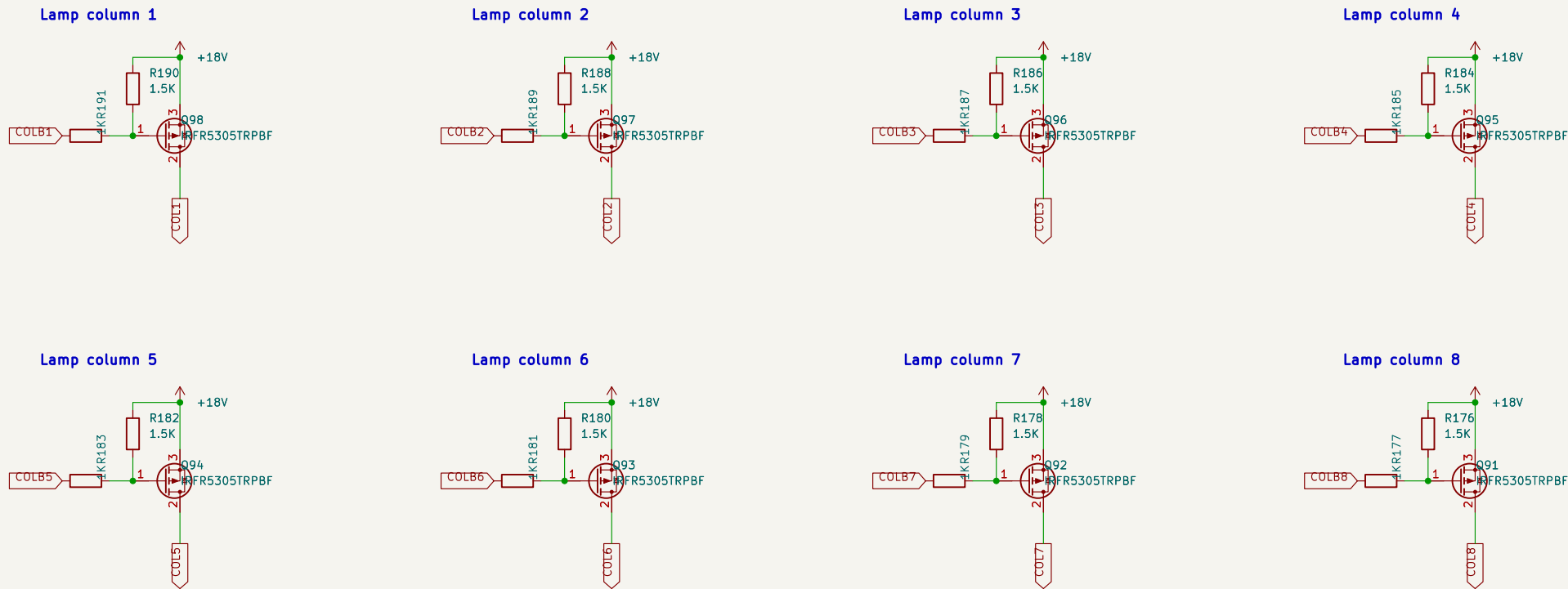
J126 Sol 17-24 drive (playfield flasher) re-bare  
J125 Sol 17-24 drive (backbox flasher) re-bare

General purpose drivers (sol. 25–28): IPD90N10S4L–06 DPAK, latch U2 output HIGH = ON, clocked by /SOL 1 (J113–12). S3M tie-back diode cathodes are brought out to J122 pins 5,6,8,9 so the harness can tie them to the supply actually used (+50V coil or +20V flasher). Outputs J122 (playfield, key pin 7), J123 (backbox, key pin 2) and J124 (cabinet, key pin 4) as on the A–12697. U2 bits 4–7 unused.





Lamp matrix column drivers. Column strobe register 0x3FE5: bit(k-1)=1 selects column k, clocked by /LMP COL (J113-8). The TBD62083A inputs have an internal pull-down (Toshiba usage note), so a tri-stated U18 (BLANKING) leaves every column OFF - the OEM 74LS374/ULN2803 pair behaves the same way. U18 latches the data bus; the TBD62083A DMOS sink pulls the P-MOSFET gate down through 1 k against the 1.5 k gate-source resistor (Vgs = -0.6 x V+18 = -8..-12 V) -> column high (+18V) with -65 mOhm (IRFR5305 DPAK; TIP107 dropped -1.2 V). Column k = Q(99-k): col1=Q98 ... col8=Q91. Outputs J137/J138 pins 1-7,9 (key pin 8).



Modern re-implementation of the Williams A-12697-3 (same form factor)

Sheet: /Lamp\_Columns/  
File: lamp\_columns.kicad\_sch

Title: Lamp matrix columns (IRFR5305 high side, +18V)

Size: A2 Date: 2026-09-04

KiCad E.D.A. 10.0.6

Rev: A

Id: 8/10



[illegible]



General illumination: each 6.3VAC string is fused (F106–F110) on the hot side and switched by a triac in the RETURN path; MT1 of the five triacs form the G\_LRET bus that goes back to the transformer on J115–7/8/10/11/12 and is referenced to logic ground by W1 (OEM: J115–1 GND REF), so the 5 x 5 A return current stays off the ground plane. Latch U1 (clocked by 7TRJAC, J113–13) output HIGH = string ON. The gate is fed by an NPN emitter follower from +5V through 27 R (-80 mA, positive gate current in quadrants I and IV -> 4-quadrant "C" grade triac; snubberless BW/CW types cannot fire in Q IV). 10 k base pull-downs keep the strings OFF while the latch is tri-stated by BLANKING. Bits 5–7 (T5–T7) go to J111 (GPIO, CPU sheet); bit 7 also drives the optional flipper relay (DNP).

G.I. string #1 (Wht–Brn / Brn)

G.I. string #2 (Wht–Org / Org)

G.I. string #3 (Wht–Yel / Yel)

G.I. string #4 (Wht–Grn / Grn)

G.I. string #5 (Wht–Vio / Vio)

